Intelligent Multi-Robot Warehouse Navigation System Using Advanced Graph Algorithms

Project Idea

The project proposes an autonomous warehouse in which multiple robots transport packages between locations without human intervention. The warehouse is represented as a graph where intersections are nodes and roads are edges.

How the System Works

1. A delivery request is created.
2. The system selects the most suitable robot for the task.
3. The robot calculates the shortest route to the destination.
4. The robot moves through warehouse intersections.
5. RFID tags placed at every node allow the robot to identify its current position.
6. If another robot is using the same node, the system prevents collisions.
7. If an obstacle blocks a route, a new route is calculated automatically.
8. Once the destination RFID tag is reached, the robot stops and completes the delivery.

Graph Algorithms Used

A* Search Algorithm – Finds the shortest route for the robot.
Graph Coloring – Prevents robots from entering the same node simultaneously.
Dynamic Shortest Path – Recalculates paths when obstacles appear.
Bipartite Matching – Assigns tasks to the most suitable robot.
Minimum Spanning Tree – Optimizes communication and infrastructure connections.

RFID Based Localization

Each warehouse intersection contains an RFID tag. Every robot carries an RFID reader below its chassis. Whenever a robot reaches an intersection, it reads the RFID value and updates its position in the graph. When the current RFID matches the destination RFID, the robot knows it has reached its destination.

Hardware Components

• ESP32 Microcontroller
• RFID Reader Module
• RFID Tags
• Motor Driver
• DC Motors and Wheels
• Ultrasonic Sensor
• Battery Pack
• Robot Chassis

Applications

Smart warehouses, logistics centers, manufacturing plants, hospital supply transport systems, and airport baggage handling systems.

Conclusion

The project combines robotics, embedded systems, RFID localization, and advanced graph algorithms to create a practical and intelligent warehouse automation system capable of real-time decision making.

Algorithm Summary

Algorithm	Purpose
A* Search	Shortest path planning
Graph Coloring	Collision avoidance
Dynamic Shortest Path	Obstacle rerouting
Bipartite Matching	Task allocation
Minimum Spanning Tree	Communication optimization